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Structure of the Thermolabile Mutant Aldolase B, A149P: Molecular Basis of Hereditary Fructose Intolerance

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Hereditary fructose intolerance (HFI) is a potentially lethal inborn error in metabolism caused by mutations in the aldolase B gene, which is critical for gluconeogenesis and fructose metabolism. The most common mutation, which accounts for 53% of HFI alleles identified worldwide, results in substitution of Pro for Ala at position 149. Structural and functional investigations of human aldolase B with the A149P substitution (AP-aldolase) have shown that the mutation leads to losses in thermal stability, quaternary structure, and activity. X-ray crystallography is used to reveal the structural basis of these perturbations. Crystals of AP-aldolase are grown at two temperatures (4 °C and 18 °C), and the structure solved to 3.0 Å resolution, using the wild-type structure as the phasing model. The structures reveal that the single residue substitution, A149P, causes molecular disorder around the site of mutation (residues 148–159), which is propagated to three adjacent β-strand and loop regions (residues 110–129, 189–199, 235–242). Disorder in the 110–129-loop region, which comprises one subunit–subunit interface, provides an explanation for the disrupted quaternary structure and thermal instability. Greater structural perturbation, particularly at a Glu189-Arg148 salt bridge in the active-site architecture, is observed in the structure determined at 18 °C, which could explain the temperature-dependent loss in activity. The disorder revealed in these structures is far greater than that predicted by homology modeling and underscores the difficulties in predicting perturbations of protein structure and function by homology modeling alone. The AP-aldolase structure reveals the molecular basis of a hereditary disease and represents one of only a few structures known for mutant proteins at the root of the thousands of other inherited disorders.

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Introduction

Hereditary fructose intolerance (HFI) is an autosomal recessive disorder that afflicts approximately one in 20,000 individuals.[1,2] In those affected, the ingestion of fructose from food or other sources leads to a multifaceted array of symptoms that may include vomiting, gastrointestinal pain, hypoglycemia, and liver and kidney failure.[2] The failure to diagnose HFI can easily be fatal due to persistent or inadvertent ingestion of fructose.[3]

HFI is caused by a deficiency in the activity of aldolase B, the isozyme of class I aldolase expressed in the liver, small intestine, and kidneys.[4,5] Among the three vertebrate aldolase isozymes (A, B, and C), aldolase B is responsible for cleavage of the metabolic intermediate fructose 1-phosphate (Fru-1-P) into dihydroxyacetone phosphate (DHAP) and glyceraldehyde in the fructose metabolism pathway. In addition, aldolase B is important for glucose homeostasis in the liver *via* the synthesis of fructose 1,6-bisphosphate (Fru-1,6-P_2) from DHAP and glyceraldehyde 3-phosphate as part of gluconeogenesis[2]. Defects in the aldolase B gene have been found to cause HFI.[6] Currently, 24

Abbreviations used: DHAP, dihydroxyacetone phosphate; Fru-1,6-P_2, fructose-1,6-bisphosphate; Fru-1-P, fructose 1-phosphate; HFI, hereditary fructose intolerance; pdb, Protein Data Bank.

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mutations in the aldolase B gene are known to be associated with HFI. The most prevalent allele encodes an Ala→Pro substitution at position 149 and accounts for bout 55% of HFI alleles identified worldwide[7]. Assuming Hardy–Weinberg equilibrium[8] of this allele in all HFI patients, approximately 80% of HFI patients carry at least one copy of A149P.

The overexpression, purification, and characterization of the aldolase B enzyme carrying the A149P mutation (AP-aldolase) shows a disruption of quaternary structure and partial activity, albeit with a marked temperature sensitivity.[9] The specific activity toward Fru-1,6-P_2 and Fru-1-P decreases from 16% to 0.5% of wild-type levels between 10 °C and 30 °C, with a $T_{1/2}$ of 25 °C. Circular dichroism experiments similarly revealed decreased stability of the tertiary and secondary protein structures of AP-aldolase, with $T_{1/2}$ values of 33 °C and 45 °C, respectively, as compared to the wild-type values of 43 °C and 50 °C. On the other hand, changes in the K_m values of AP-aldolase with respect to the wild-type enzyme were temperature-independent, but showed considerably greater effects for the substrates DHAP and Fru-1-P, suggesting a structural perturbation at the C1-phosphate binding site of AP-aldolase.[9] Further evidence shows some link between the temperature-sensitive loss of activity and the loss of quaternary structure. In contrast to the tetrameric wild-type aldolase B, AP-aldolase forms dimers in solution from 4 °C to 30 °C, with some dissociation into monomeric species above 30 °C.[9] This finding is in agreement with the canonical role of the quaternary structure of aldolase in promoting structural stability.[10] Interestingly, several other mutations of aldolase B associated with HFI,[11,12] as well as mutations in the aldolase A isozyme associated with a rare form of hemolytic anemia,[13,14] have been shown to affect quaternary structure.

Although the structure of >1000 proteins involved in biological pathways have been determined, only a handful are representative of dysfunctional proteins resulting from genetic mutations.[15–18] This is due to the inherent difficulty of structural analysis of unstable proteins. Here, the structure of one such protein, AP-aldolase, is determined by X-ray crystallography at 4 °C and 18 °C, revealing the molecular basis of the structural perturbations and loss of activity with increased temperature. The structures showed that the point mutation causes extensive structural perturbation both at the site of substitution and in adjacent distal loop regions, providing important insights into the mutant phenotype. The extent of disorder is reflected in the model statistics, as has been reported for other unstable proteins.[19] This study lends insight into the molecular processes of disease pathogenesis and gives a view of the utility of modeling studies in understanding the molecular perturbations inherent in "mutant" proteins.

Results

Overall structure

AP-aldolase is a thermolabile form of aldolase B, which exhibits loss in secondary and tertiary structure at 45 °C and 33 °C, respectively. However, catalytic activity is lost at >15 °C that is not accounted for by changes in the CD spectra.[9] The X-ray crystal structure of AP-aldolase was solved at 4 °C and 18 °C in order to visualize the structural changes occurring at a temperature where the decline in catalytic activity is most acute. It should be noted that the crystals grown at 4 °C were transferred into cryo conditions at 4 °C and frozen in liquid nitrogen to prevent any exposure to higher temperatures during mounting on the X-ray beam.

The AP-aldolase crystals showed relatively weak diffraction, although improved resolution was obtained from larger crystals. In addition to the data presented here, three to four complete datasets of AP-aldolase were collected at each temperature. Overall anisotropy in the diffraction pattern was always observed, regardless of the crystal growth temperature or cryoprotectant conditions. The data were consistently weakest along the *l*-axis, with a resolution limit for strong reflections of about 4 Å on that axis, while the *h* and *k*-axes had resolution limits of 3 Å or better. The anisotropic alignment of the reflections along the crystallographic axes signifies that it is an inherent property of the crystal form, and not merely due to stochastic factors. In addition, this interpretation is fully consistent with the anisotropic distribution of the *B*-factors. In the structure of AP-aldolase at 18 °C, for example, the overall anisotropic temperature factors (B_{11} = 16.20 Å^2, B_{22} = 12.39 Å^2, and B_{33} = −28.59 Å^2), together with the average *B*-factor for the structure (60.3 Å^2), signifies that the effective average *B*-factors in the *x*, *y*, and *z* directions are 44.1 Å^2, 47.91 Å^2, and 88.89 Å^2, respectively. It should be noted that the scaling together or merging of separate datasets failed to improve completeness or dataset statistics.

In both AP-aldolase structures, local effects of the substitution and propagated effects on structural order were observed, as reflected by the relatively high overall *B*-factors. Similar disorder was observed in a recent study on the structure of glutathione reductase from *Plasmodium falciparum*.[19] In this investigation, the structure was solved to 2.6 Å resolution by molecular replacement, with the final model having R_{work} = 25.1%, R_{free} = 30.2%, and an average protein *B*-factor of 80 Å^2. The structure had areas of missing density in three different regions of the protein, corresponding to 27, 32, and four residues (out of a total of 478 residues in the monomer), similar to observations reported here. The authors argued that the model with the disordered regions was correct based on observations from annealed omit maps, anisotropy of the crystals along one crystal axis, a reasonable *R*-factor distribution *versus* resolution, and reasonable

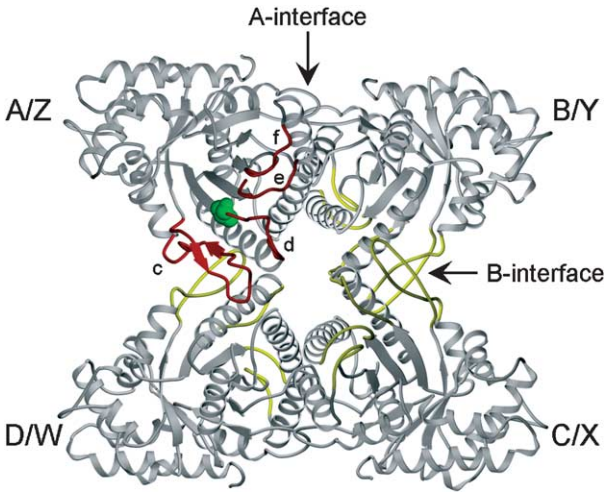


Figure 1. Orientation of the AP-aldolase monomers. The relative orientation of the aldolase tetramer with each of eight monomers (A–D, W–Z) in the asymmetric unit is indicated. The four loops from the wild-type aldolase B structure (pdb 1QO5), which are disordered in the AP-aldolase structure, are depicted in monomer A/Z (red, labeled c–f). The same loops are in yellow in the remaining subunits. The substituted proline Pro149 is shown as cpk (*green*) and the *A* and *B*-interfaces are indicated by arrows.

model geometry. A similar fraction of the protein was disordered in the AP-aldolase structures with comparable effects on data collection and model statistics, including *R* and *B*-factors.

At the two temperatures, AP-aldolase crystals were obtained under similar conditions and in the same space group ($P2_12_12$), with nearly identical unit-cell dimensions (see Experimental Procedures). At both temperatures, eight monomers of AP-aldolase were present in the asymmetric unit, arranged as two tetramers (A, B, C, D and W, X, Y, Z), in an orientation similar to that of the wild-type aldolase B tetramer. Dimers A and B, C and D, W and X, and Y and Z interact *via* the helix-rich *A*-interface, and dimers A and D, B and C, W and Z, and X and Y *via* the loop–loop *B*-interface (Figure 1). For the 4 °C structure, the final *R*-factor was 30.4% (R_{free} = 34.9%) for all data in resolution range 20.2–3.0 Å. For the 18 °C structure, the final *R*-factor was 27.1% (R_{free} = 31.9%) for all data in resolution range 79.4–3.0 Å. The quality of the structures was analyzed using the program PROCHECK[20] showing 79.7% of residues in the core region and 20.3% in the allowed region for the 4 °C structure and 81.5% of residues in the core region and 18.5% in the allowed region for the 18 °C structure. The coordinate error as calculated by Luzzati plots was 0.55 Å and 0.44 Å for the 4 °C and 18 °C structures, respectively.

At both temperatures, there were four regions with weak or missing density in most monomers (Table 1). These disordered regions were all clustered around the site of substitution. There was no connection between the location of the missing regions in the structure and crystal contacts. Therefore, the missing regions in the structure are not a consequence of the crystal quality *per se*, but instead likely reflect molecular disorder caused by perturbations at the site of the substitution that are propagated to the adjacent regions in the protein (i.e. distal effects).

The superposition of either AP-aldolase with the wild-type aldolase B structure (1QO5, monomer B[21]) indicated that the overall protein architecture was unchanged. The average rmsd between the α-carbons of the 4 °C monomers and wild-type was 0.74 Å, whereas with the 18 °C monomers it was 0.87 Å. An overlay of the active-site residues of the two AP-aldolase structures with that of wild-type aldolase B is shown in Figure 2, demonstrating that

Table 1. Disordered loop regions in the AP-aldolase structures

Monomer	Loop c[a]	Loop d	Loop e	Loop f
4 °C structure				
A	110–124	153–158	193–198	235–242
B	110–125	149–158	190–199	236–242
C	110–113		193–194	236–239
D	110–125	149–158	191–199	235–242
W	110–125	149–158	189–196	235–242
X	111–114			236–242
Y	110–125	151–158	190–197	236–242
Z	110–125	154–159	191–196	235–242
18 °C structure				
A	110–125	150–158	192–194	236–242
B	110–127	150–158	194–196	235–240
C	110–129	148–158	193–196	235–240
D	110–127	150–157	190–196	235–242
W	109–128	150–157	191–195	235–242
X	110–129	148–158	193–196	236–237
Y	110–127	150–158	194–196	235–240
Z	110–127	149–158	190–196	235–242

Numbers denote the extent (given as residue numbers) of each disordered loop region. The loop names are based on the nomenclature used for aldolase A by Pan & Smith.[49]

[a] Loop identification depicted in Figure 1.

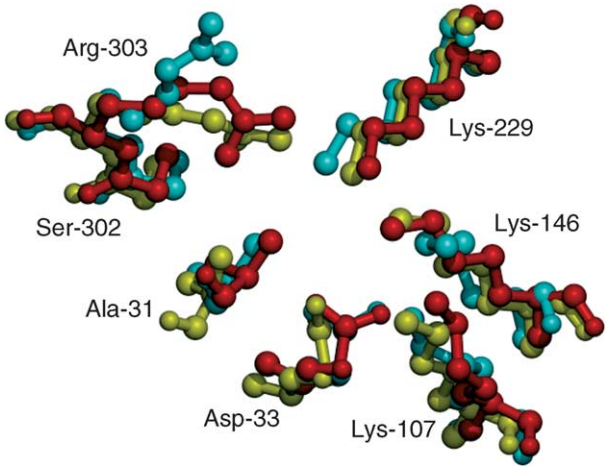


Figure 2. Overlay of the active sites of AP-aldolase and wild-type aldolase B. Overlay of the active-site residues are depicted for wild-type aldolase B (cyan, pdb 1QO5, chain B), AP-aldolase structure at 4 °C (yellow, monomer X), and AP-aldolase structure at 18 °C (red, monomer B).

the orientation of active-site residues of AP-aldolase was intact at the two temperatures. The exception was the side-chain of Arg303, which the electron density clearly showed was oriented closer to Lys229 (5.8 Å compared to 9.4 Å in wild-type) in all eight monomers at both temperatures. Arg303 is a conserved residue in the aldolase active site and may be important in the catalytic mechanism. The aldolase mechanism proceeds *via* binding hexose, Schiff-base formation, C3–C4 bond cleavage, release of the first triose, Schiff-base hydrolysis, and finally, release of the second triose.[22] Arg303 has been shown to have alternative capacities in binding to the C6-phosphate of the linear hexose substrate,[23] or the C1-phosphate of the covalently bound product,[24] or to a residue near the C terminus (Glu354) in an enzyme–product complex.[25] Thus Arg303 switches roles between binding the substrate/product and binding the C terminus. Given that the rate-limiting step is product release[26,27] and likely involves a conformational change in the C terminus, this succession of liganded structures indicated that Arg303 may be a "trigger" residue in modulating the action of the C terminus during catalysis.[24,28] This new conformation of Arg303 in the AP-aldolase structures could explain the >tenfold loss in activity at all temperatures.[9] The electron density shown in Figure 3 around Arg303 and other active-site residues from monomer W was typical for all 16 monomers.

AP-aldolase structure at 4 °C

The electron density maps of the AP-aldolase structure solved at 4 °C, where AP-aldolase has maximal activity,[9] had four loop regions of weak or missing density in six of eight monomers. Also missing were the first residues at the N terminus and the last 18–21 residues of the C-terminal region, which are not visible in most class I-aldolase structures.[21,23,29,30] The four disordered regions (loops c, d, e, and f; see Figure 1 and Table 1) are localized at the C-terminal end of the β-strand containing the site of the substitution and in the loop immediately following (residues 149–158), as well as in two flanking regions (residues 190–197 and 236–242) and the large region from residues 110–125. Monomers X and C were the exception, with complete density visible at the site of substitution (residues 149–158; loop d) as well as for most or all residues in loops c and e (Table 1). All three loops had electron density corresponding to the wild-type aldolase B structure, showing that AP-aldolase at 4 °C can adopt a conformation similar to that of wild-type (Figure 4(a)). An examination of crystal contacts showed that monomers X and C are not stabilized to any greater extent than the other monomers by packing interactions. Poor electron density was visible at some of the regions where a model was not built. For example, in monomer D repeated attempts to build a model for residues 115–119 and 121–125 resulted in increased *R*-factors upon refinement.

In the regions where density was visible, the structure of AP-aldolase was very similar to that of wild-type aldolase B. The missing regions were likely a consequence of molecular disorder in the loops including and flanking the substituted

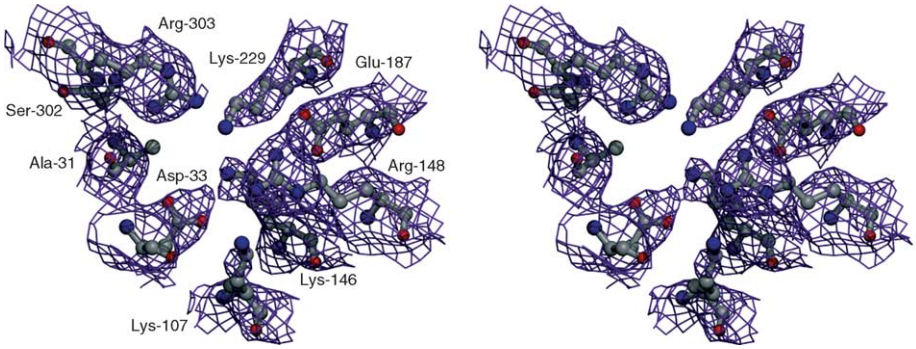


Figure 3. Stereo view of the active site. The 18 °C AP-aldolase structure (monomer W) is depicted with the $2F_o - F_c$ map contoured at a 1.0σ.

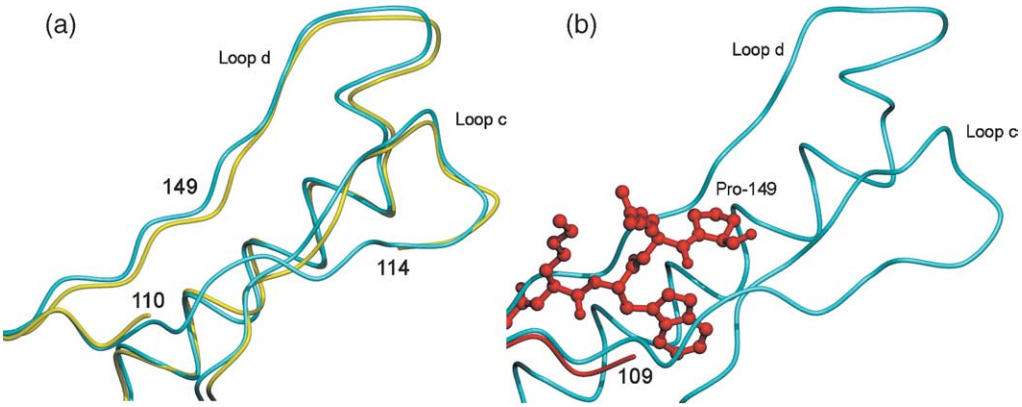


Figure 4. The conformation of AP-aldolase at the site of substitution and in the adjacent loops c and d. (a) Overlay of the AP-aldolase structure at 4 °C (yellow, monomer X) with the wild-type model (cyan). The site of the 149-substitution is indicated and residues flanking the disordered regions are numbered. (b) Overlay of the AP-aldolase structure at 18 °C (red, monomer B) with the wild-type model (cyan). Residues 146–149 for AP-aldolase are displayed as ball-and-stick and loops are labeled as in (a).

residue, which is consistent with the high group *B*-factors of residues at the margins of these regions.

Structure of AP-aldolase at 18 °C

The structure of AP-aldolase at 18 °C corresponds to the temperature at which the activity begins to decline as the enzyme approaches physiologic temperature.[9] As in the 4 °C structure, the structure of AP-aldolase at 18 °C displayed disorder, characterized by areas of missing density in the four loops listed in Table 1. As in the 4 °C structure, residues at the N and C termini were missing from the model. In contrast to the 4 °C structure, however, none of the loops showed complete density in any of the eight monomers. Where there was clear density for residues 148–150, the structure deviated into the region occupied by residues 123–125 in the wild-type structure, the assignment of this density to the 149 chain *versus* the 123 chain was ambiguous, so no model was included after residue 149 (Figure 4(b)). For all eight monomers, residues just beyond proline 149 (150–157) were disordered, showing little or no density in the maps. When the AP-aldolase structures were superimposed on the wild-type structures, the 18 °C structure revealed a greater deviation than for the 4 °C structure near the disordered regions (Figure 5). The superimposed α-carbons of residues adjacent to the four disordered loops had an average rmsd of 1.7 Å (ranged from 0.6 Å to 4 Å) for the structure at 18 °C and 0.79 Å (ranged from 0.5 Å to 1.3 Å) for the structure at 4 °C.

A key distinction between the AP-aldolase structure at 18 °C and that of all wild-type aldolase isozymes[21,29,30] was the finding that Glu189 no longer makes a salt bridge with Arg148. This conserved salt bridge, which is part of the active-site architecture,[23] is shown for wild-type aldolase B in Figure 6. In particular, Glu189 in all eight

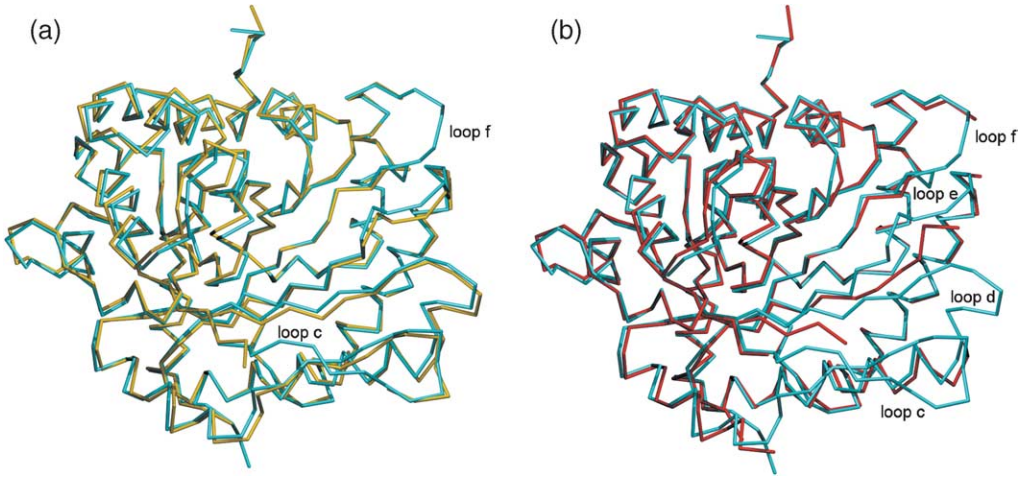


Figure 5. Comparison of AP-aldolase structures with wild-type structure. (a) Overlay of the α-carbon trace of the AP-aldolase structure at 4 °C (yellow, monomer X) with that of the wild-type model (cyan). (b) Overlay of the α-carbon trace of the AP-aldolase structure at 18 °C (red, monomer B) with that of the wild-type model (cyan). Disordered loops in AP-aldolases are noted on the wild-type trace.

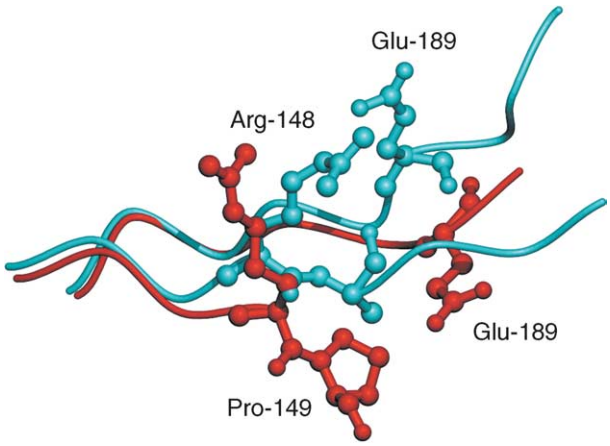


Figure 6. Loss of a conserved salt-bridge at active site in AP-aldolase. Overlay of the AP-aldolase structure at 18 °C (red, monomer W) with the wild-type model (cyan) rendered as coils with the Glu189/Arg148 salt-bridge and residue 149 rendered as ball-and stick.

monomers of the 18 °C structure was oriented in the opposite direction from that in the wild-type structure. Glu189 appeared to make no contacts in this swung-out position, however, the former partner, Arg148, was within 3.5–3.9 Å of Glu187 in monomers A and W of the 18 °C structure. In the other monomers Arg148 was either not in the model or was >4.3 Å from Glu187, sometimes with an intervening water molecule. Glu187 is important in the mechanism of Schiff-base chemistry acting as a general acid catalyst in the dehydration of the carbinolamine intermediate[24,31] (Choi, K. H., K.N.A. & D.R.T., unpublished results). A new salt bridge formed with Glu187 could interfere with the normal catalytic mechanism by hindering acid–base chemistry. Whereas the absence of the Glu189/Arg148 salt bridge was found in all monomers in the structure at 18 °C, only monomers B and Y showed this loss in the 4 °C structure. Thus, the altered conformation of Glu189 may be a likely explanation for the temperature-dependent decline in activity at temperatures above 15 °C.

Discussion

Frequently the instability of mutant proteins implicated in human disease discourages empirical experimental structure/function studies. Homology modeling is used to map the consequences of deleterious mutations onto structures of native or structurally related proteins in order to predict the structural and functional consequences of the mutation.[32–34] However, the necessity of using a modeling approach by its nature makes validation difficult without sufficient empirical structural studies on similar systems. In one study, a theoretical modeling program was demonstrated to be no better at predicting deleterious effects of missense mutations than a program that made predictions based solely on conservation of primary structure among homologous proteins.[35] The utility of structure determination of mutant proteins is exemplified by the study of the hemoglobin structure in sickle-cell anemia.[36] Despite this fact, only a handful of structures have been determined for proteins from missense mutations involved in human diseases.[16–18] Thus, we have undertaken the structure of unstable mutant proteins involved in HFI to assess the extent of structural changes and their relationship to the loss of function.

Structural consequences of the Ala→Pro substitution in AP-aldolase

The unique nature of a proline substitution imposes severe constraints on the polypeptide backbone. In AP-aldolase the substitution of proline clearly disrupts the native β-strand structure at position 149. In addition, this perturbation results in a change in structure at positions that play a role in catalysis, i.e. Arg303 and Glu189/Arg148, and there are distal effects imposed on the loop regions flanking both sides of the substitution. The deviation of loop d into the region occupied by loop c leads to disorder in loop c and disrupts the conformation necessary for oligomerization *via* the *B*-interface.[10] In addition, the shift at the site of substitution causes changes in loops e and f. Thus, the structure of AP-aldolase demonstrates that the single missense mutation causes extensive distal effects to the protein structure. Previous studies have shown that the quaternary structure of AP-aldolase is disrupted, adopting a dimeric state in solution, instead of the wild-type tetramer.[9] It is likely that the thermolability displayed by the mutant enzyme is a direct consequence of its inability to properly oligomerize.[10,13,14,37]

Effects on catalysis

Previous investigations have shown that purified AP-aldolase exhibits low levels of catalytic activity toward the reactants normally utilized by aldolase *in vivo*. The residual activity is extremely thermolabile, with a $T_{1/2}$ of 25 °C and a peak k_{cat} corresponding to 16% of wild-type activity at a stable temperature of 10 °C.[9] A comparison of the AP-aldolase structures with wild-type aldolase suggests that changes in three active-site residues, Arg148, Glu189, and Arg303, could account for the k_{cat} decrease. Arg303 is a multifunctional residue that has been hypothesized to undergo a conformational switch during catalysis,[24,28] and could be involved in the release of DHAP, the rate-limiting step for both aldolase A[27,38] and aldolase B (Choi, K. H. & D.R.T., unpublished results). Mutations at this site (R303A in aldolase A and R303W and R303Q in aldolase B) demonstrate an effect on both the k_{cat} and K_m values.[23,39,40] In wild-type aldolase, Arg148 and Glu189 form a conserved salt bridge that constitutes one surface of a cleft leading to the

active site.[21,24,29] Mutations at either of these two residues in aldolase A (E189A and R148A) lead to a 70–300-fold decrease in k_{cat} value.[41] The change in the orientation of Glu189 seen predominantly in the 18 °C AP-aldolase structure, which was caused by the perturbation of loop e adjacent to the site of substitution, was consistent with the decreased catalytic activity. The disordered loop regions and consequences near the active site were more extensive at 18 °C compared to those at 4 °C. Thus, the structural perturbation associated with a local conformational change explains the temperature dependence of the AP-aldolase structure.

Insights into the viability of homology modeling as a predictive tool

The use of homology modeling for AP-aldolase, along with many other substitution mutations that have been found as the cause of genetic diseases, have been reviewed.[42] The result of homology modeling predicted that the A149P mutation leads to backbone strain in the β-strand, possibly causing protein instability, with no prediction of any propagated effects. The findings from the AP-aldolase structure raise general issues regarding the validity of using homology modeling as a predictive tool for the deleterious effects of mutations on protein function. The AP-aldolase structures showed "distal effects" of the single-residue substitution, in that three regions (loops c, e, and f) aside from the site of substitution are significantly affected in the mutant structure. The limitations of a theoretical modeling approach were especially underscored by the changes observed in the quaternary structure of AP-aldolase. Even with *a priori* knowledge of a quaternary structural perturbation, a theoretical approach would not have predicted with any certainty which of the subunit interfaces were actually disrupted, since they are equidistant from the site of the substitution.

X-ray crystallography can reveal the structural consequences of disease-causing mutations that cannot easily be anticipated from homology modeling. This study will provide a basis for improving such theoretical predictions. Herein, the structure of AP-aldolase was solved at two temperatures, giving a detailed picture of the temperature-dependent loss of structure. This includes an altered conformation for a conserved salt-bridge at the active site explaining the loss in activity with increased temperature. The disordered regions correspond to loops involved in the dimer–dimer interface in the wild-type tetramer, thus revealing the basis for the known loss in quaternary structure of AP-aldolase.[9] The prevalence of the A149P mutation in HFI makes AP-aldolase a very good potential target for therapies that would aim to restore stability and functionality to the thermolabile enzyme using small molecule ligands. The structure makes possible the search for such ligands *via* screening of chemical libraries in combination with structure-based ligand design.

Experimental Procedures

Purification and crystallization

AP-aldolase was produced by heterologous expression in *Escherichia coli* DH5α as a glutathione-*S*-transferase-fusion protein and subsequent cleavage to yield the full length, purified, active protein, as described[9]. The crystals were obtained by the vapor diffusion method with hanging-drop geometry, by mixing equal volumes of 20 mg/ml purified AP-aldolase in 1 mM phosphate-buffered saline (pH 7.4), 1 mM DTT, with 1.8–2.2 M $(NH_4)_2SO_4$ and 2% (w/v) 1,8-diaminooctane at 4 °C and equilibrating over wells containing the same buffer. For the crystals grown at 18 °C, crystallization trays were transferred to 18 °C after setup and overnight incubation at 4 °C. Crystals appeared within one to two weeks.

Data collection

Data for the 4 °C crystals were collected on beamline X4A at the National Synchrotron Light Source at Brookhaven National Laboratories. Cryoprotection conditions consisted of successive transfers of single crystals at 4 °C into a drop of well solution containing 15% (w/v) glucose for one minute, and into a drop of well solution containing 30% glucose plus 5% (v/v) glycerol for 30 seconds. The crystals were frozen in liquid nitrogen and transferred to a stream of nitrogen gas cooled to 93 K. A full dataset was collected to 2.6 Å using a quad4 detector, with 0.5° oscillations for 180° total, and a crystal-to-detector distance of 200 mm. The Denzo and Scalepack packages[43] were used for data indexing, reduction, and scaling. The space group was determined to be $P2_12_12$, with unit cell dimensions of $a=153.48$ Å, $b=153.51$ Å, $c=186.48$ Å.

Prior to data collection, the 18 °C crystals were dehydrated[44,45] at the same temperature by vapor diffusion against 3.5 M $(NH_4)_2SO_4$ for three hours. Paratone-N was used as cryoprotectant and the crystals were frozen as for the 4 °C crystals. Data were collected at the Boston University School of Medicine on a Rigaku RU300 X-ray generator equipped with an R-Axis IV^{++} area detector. A full dataset was collected at 2.7 Å with 1° oscillations for 180° total, and a crystal-to-detector distance of 200 mm. The space group was determined to be $P2_12_12$, with unit cell dimensions of $a=153.42$ Å, $b=153.48$ Å, $c=185.55$ Å. Due to the high R_{sym} values in the outer shell, both data sets were refined to 3.0 Å.

Molecular replacement

The AP-aldolase structure was solved by molecular replacement using the program AMoRe[46] and the structure of wild-type aldolase B (pdb *1QO5*;[21]) as the initial search model. The best search model consisted of a dimer, as opposed to the tetramer (the A–B dimer[10]). Eight monomers were identified in the asymmetric unit (designated A–D and W–Z), arranged as two tetramers, consistent with the wild-type oligomeric structure. Orientation of the eight monomers obtained by molecular replacement was improved by rigid-body refinement using CNS.[47]

Structural refinement

The SigmaA-weighted $2F_o-F_c$ maps, calculated from the model after rigid-body refinement, revealed regions

Table 2. Data collection and refinement statistics

	4 °C model	18 °C model
Data collection		
Resolution (Å)	∞–3.0	∞–3.0
Total no. of reflections	410,514	371,291
No. of unique reflections	91,423	83,595
Completeness (overall/outer shell[a]) (%)	94.9/90.2	95.1/88.8
R_{sym} (overall/outer shell)	0.061/0.283	0.067/0.364
I/σ (overall/outer shell)	18.7/2.6	12.1/2.1
Refinement statistics		
No. of protein atoms (non-hydrogen)	18,879	18,483
No. of water molecules	21	35
No. of sulfate molecules	5	8
Resolution (Å)	20.2–3.0	79.4–3.0
σ cutoff	2.0	2.0
R_{cryst}[b]	30.4	27.1
R_{free}[c]	34.9	31.9
Mean *B*-factor (Å^2)	64.9	60.3

Data were collected to higher resolution (2.6 Å and 2.7 Å, respectively) but data refer to that with a 3.0 Å cutoff.
[a] Values for the outermost shell are for the range 3.11–3.0 Å for both the 4 °C and 18 °C structures.
[b] $R=\Sigma[(I-\langle I\rangle)^2]/\Sigma I^2$.
[c] R_{free} was calculated from a random selection constituting 10% of the data.

of missing or ambiguous electron density in most monomers (see Results). These regions were excluded from the model and built in, where applicable, during rounds of refinement using $2F_o-F_c$ simulated-annealing omit maps. Iterative rounds of model building were done using O[48] and positional, group temperature-factor, and simulated annealing refinement procedures in the program CNS. For refinement, a test set consisting of 10% of the data was used for R_{free} calculations. Tight NCS restraints (weight=300) were applied between pairs of monomers in the asymmetric unit (A=Z; B=Y; C=X; D=W), based on the similarity of the refined group *B*-factor values. In the final rounds of refinement, $2F_o-F_c$ maps were used to include water molecules using a 3σ cutoff. Several sulfate molecules were also identified in the active site. The data collection, refinement, and model statistics are summarized in Table 2.

Protein Data Bank accession codes

The coordinates of the refined structures have been deposited in the Protein Data Bank with accession codes 1XCL and 1XCM.

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References

1. James, C. L., Rellos, P., Ali, M., Heeley, A. F. & Cox, T. M. (1996). Neonatal screening for hereditary fructose intolerance: frequency of the most common mutant aldolase B allele (A149P) in the British population. *J. Med. Genet.* **33**, 837–841.
2. Steinmann, B., Gitzelmann, R. & Van den Berghe, G. (2001). Disorders of Fructose Metabolism. In *The Metabolic and Molecular Basis of Inherited Disease* (Scriver, C., Beaudet, A., Sly, W. & Valle, D., eds), vol. 1, pp. 1489–1520, McGraw-Hill, Inc, New York.
3. Ali, M., Rosien, U. & Cox, T. M. (1993). DNA diagnosis of fatal fructose intolerance from archival tissue. *Quart. J. Med.* **86**, 25–30.
4. Lebherz, H. G. & Rutter, W. J. (1969). Distribution of fructose diphosphate aldolase variants in biological systems. *Biochemistry*, **8**, 109–121.
5. Rutter, W. J., Blostein, R. E., Woodfin, B. M. & Weber, C. S. (1963). Enzyme variants and metabolic diversification. *Adv. Enzyme Regul.* **17**, 39–56.
6. Hers, H.-G. & Joassin, G. (1961). Anomalie de l'aldolase hepatique dans l'intolerance au fructose. *Enzymol. Biol. Clin.* **1**, 4–14.
7. Tolan, D. R. (1995). Molecular basis of hereditary fructose intolerance: mutations and polymorphisms in the human aldolase B gene. *Hum. Mutat.* **6**, 210–218.
8. Tamarin, R. & Leavitt, R. W. (1991). *Principles of Genetics* (3rd edit.), Wm. C. Brown Publishers, Dubuque, IA.
9. Malay, A. D., Procious, S. L. & Tolan, D. R. (2002). The temperature dependence of activity and structure for the most prevalent mutant aldolase B associated with hereditary fructose intolerance. *Arch. Biochem. Biophys.* **408**, 295–304.
10. Beernink, P. T. & Tolan, D. R. (1996). Disruption of the aldolase A tetramer into catalytically active monomers. *Proc. Natl Acad. Sci. USA*, **93**, 5374–5379.
11. Rellos, P., Sygusch, J. & Cox, T. M. (2000). Expression, purification, and characterization of natural mutants of human aldolase B. Role of quaternary structure in catalysis. *J. Biol. Chem.* **275**, 1145–1151.
12. Esposito, G., Vitagliano, L., Santamaria, R., Viola, A., Zagari, A. & Salvatore, F. (2002). Structural and functional analysis of aldolase B mutants related to hereditary fructose intolerance. *FEBS Letters*, **531**, 152–156.
13. Kishi, H., Mukai, T., Hirono, A., Fujii, H., Miwa, S. & Hori, K. (1987). Human aldolase A deficiency associated with a hemolytic anemia: thermolabile aldolase due to a single base mutation. *Proc. Natl Acad. Sci. USA*, **84**, 8623–8627.

14. Beernink, P. T. & Tolan, D. R. (1994). Subunit interface mutants of rabbit muscle aldolase form active dimers. *Protein Sci.* **3**, 1383–1391.
15. Padlan, E. A. & Love, W. E. (1985). Refined crystal structure of deoxyhemoglobin S.I. Restrained least-squares refinement at 3.0-Å resolution. *J. Biol. Chem.* **260**, 8272–8279.
16. Phillips, J. D., Parker, T. L., Schubert, H. L., Whitby, F. G., Hill, C. P. & Kushner, J. P. (2001). Functional consequences of naturally occurring mutations in human uroporphyrinogen decarboxylase. *Blood*, **98**, 3179–3185.
17. Hamilton, J. A., Steinrauf, L. K., Liepnieks, J., Benson, M. D., Holmgren, G., Sandgren, O. & Steen, L. (1992). Alteration in molecular structure which results in disease: the Met-30 variant of human plasma transthyretin. *Biochim. Biophys. Acta*, **1139**, 9–16.
18. Cardoso, R. M., Thayer, M. M., DiDonato, M., Lo, T. P., Bruns, C. K., Getzoff, E. D. & Tainer, J. A. (2002). Insights into Lou Gehrig's disease from the structure and instability of the A4V mutant of human Cu,Zn superoxide dismutase. *J. Mol. Biol.* **324**, 247–256.
19. Sarma, G. N., Savvides, S. N., Becker, K., Schirmer, M., Schirmer, R. H. & Karplus, P. A. (2003). Glutathione reductase of the malarial parasite *Plasmodium falciparum*: crystal structure and inhibitor development. *J. Mol. Biol.* **328**, 893–907.
20. Laskowski, R. A., MacArthur, M. W., Moss, D. S. & Thornton, J. M. (1993). PROCHECK: a program to check the sterochemical quality of protein structures. *J. Appl. Crystallog.* **26**, 283–291.
21. Dalby, A., Tolan, D. R. & Littlechild, J. A. (2001). Crystal structure of human liver fructose 1,6-bisphosphate aldolase. *Acta Crystallog. sect. D*, **57**, 1526–1533.
22. Rose, I. A., O'Connell, E. L. & Mehler, A. H. (1965). Mechanism of the aldolase reaction. *J. Biol. Chem.* **240**, 1758–1765.
23. Choi, K. H., Mazurkie, A. S., Morris, A. J., Utheza, D., Tolan, D. R. & Allen, K. N. (1999). Structure of a fructose-1,6-bis(phosphate) aldolase liganded to its natural substrate in a cleavage-defective mutant at 2.3 Å. *Biochemistry*, **38**, 12655–12664.
24. Choi, K. H., Shi, J., Hopkins, C. E., Tolan, D. R. & Allen, K. N. (2001). Snapshots of catalysis: the structure of fructose-1,6-(bis)phosphate aldolase covalently bound to the substrate dihydroxyacetone phosphate. *Biochemistry*, **40**, 13868–13875.
25. Blom, N. & Sygusch, J. (1997). Product binding and role of the C-terminal region in class I D-fructose 1,6-bisphosphate aldolase. *Nature Struct. Biol.* **4**, 36–39.
26. Morris, A. J., Davenport, R. C. & Tolan, D. R. (1996). A lysine to arginine substitution at position 146 of rabbit aldolase A shifts the rate-determining step to Schiff base formation. *Protein Eng.* **9**, 61–67.
27. Rose, I. A., Warms, J. V. B. & Kuo, D. J. (1987). Concentration and partitioning of intermediates in the fructose bisphosphate aldolase reaction. Comparison of the muscle and liver enzymes. *J. Biol. Chem.* **262**, 692–701.
28. Dalby, A., Dauter, Z. & Littlechild, J. A. (1999). Crystal structure of human muscle aldolase complexed with fructose 1,6-bisphosphate: mechanistic implications. *Protein Sci.* **8**, 291–297.
29. Sygusch, J., Beaudry, D. & Allaire, M. (1987). Molecular architecture of rabbit skeletal muscle aldolase at 2.7-Å resolution. *Proc. Natl Acad. Sci. USA*, **84**, 7846–7850.
30. Arakaki, T. L., Pezza, J. A., Cronin, M. A., Hopkins, C. E., Zimmer, D. B., Tolan, D. R., T. & Allen, K. N. (2004). Structure of human brain fructose 1,6-bisphosphate aldolase: linking isozyme structure with function. *Protein Sci.* **13**, 3077–3084.
31. Maurady, A., Zdanov, A., de Moissac, D., Beaudry, D. & Sygusch, J. (2002). A conserved glutamate residue exhibits multifunctional catalytic roles in D-fructose-1,6-bisphosphate aldolases. *J. Biol. Chem.* **277**, 9474–9483.
32. Erlandsen, H. & Stevens, R. C. (1999). The structural basis of phenylketonuria. *Mol. Genet. Metab.* **68**, 103–125.
33. Waters, P. J. (2003). How PAH gene mutations cause hyper-phenylalaninemia and why mechanism matters: insights from *in vitro* expression. *Hum. Mutat.* **21**, 357–369.
34. Wedekind, J. E., Frey, P. A. & Rayment, I. (1995). Three-dimensional structure of galactose-1-phosphate uridylyltransferase from *Escherichia coli* at 1.8 Å resolution. *Biochemistry*, **34**, 11049–11061.
35. Ng, P. C. & Henikoff, S. (2002). Accounting for human polymorphisms predicted to affect protein function. *Genome Res.* **12**, 436–446.
36. Harrington, D. J., Adachi, K. & Royer, W. E., Jr (1997). The high resolution crystal structure of deoxyhemoglobin S. *J. Mol. Biol.* **272**, 398–407.
37. Tolan, D. R., Schuler, B., Beernink, P. T. & Jaenicke, R. (2003). Thermodynamic analysis of the dissociation of the aldolase tetramer substituted at one or both of the subunit interfaces. *Biol. Chem.* **384**, 1463–1471.
38. Morris, A. J. & Tolan, D. R. (1994). Lysine-146 of rabbit muscle aldolase is essential for cleavage and condensation of the C3–C4 bond of fructose 1,6-bis(phosphate). *Biochemistry*, **33**, 12291–12297.
39. Santamaria, R., Esposito, G., Vitagliano, L., Race, V., Paglionico, I., Zancan, L. *et al*. (2000). Functional and molecular modelling studies of two hereditary fructose intolerance-causing mutations at arginine 303 in human liver aldolase. *Biochem. J.* **350**, 823–828.
40. Santamaria, R., Tamasi, S., Del Piano, G., Sebastio, G., Andria, G., Borrone, C. *et al*. (1996). Molecular basis of hereditary fructose intolerance in Italy: identification of two novel mutations in the aldolase B gene. *J. Med. Genet.* **33**, 786–788.
41. Morris, A. J. (1995). A Proposed Catalytic Mechanism of Rabbit Aldolase A Based Upon Site-Directed Mutagenesis. PhD Dissertation, Boston University.
42. Wang, Z. & Moult, J. (2001). SNPs, protein structure, and disease. *Hum. Mutat.* **17**, 263–270.
43. Otwinowski, Z. & Minor, W. (1997). Processing of X-ray diffraction data collected in oscillation mode. *Methods Enzymol.* **276**, 307–326.
44. Kuo, A., Bowler, M. W., Zimmer, J., Antcliff, J. F. & Doyle, D. A. (2003). Increasing the diffraction limit and internal order of a membrane protein crystal by dehydration. *J. Struct. Biol.* **141**, 97–102.
45. Heras, B., Edeling, M. A., Byriel, K. A., Jones, A., Raina, S. & Martin, J. L. (2003). Dehydration converts DsbG crystal diffraction from low to high resolution. *Structure*, **11**, 139–145.
46. Navaza, J. (1994). AMoRe: an automated package for molecular replacement. *Acta Crystallog. sect. A*, **50**, 157–163.
47. Brunger, A. T., Adams, P. D., Clore, G. M., DeLano, W. L., Gros, P., Grosse-Kunstleve, R. W. *et al*. (1998). Crystallography & NMR system: A new software suite for macromolecular structure determination. *Acta Crystallog. sect. D*, **54**, 905–921.
48. Jones, T. A., Zou, J. Y., Cowan, S. W. & Kjeldgaard, M. (1991). Improved methods for building protein

models in electron density maps and the location of errors in these methods. *Acta Crystallog. sect. A*, **47**, 110–119.

49. Pan, H. & Smith, D. L. (2003). Quaternary structure of aldolase leads to differences in its folding and unfolding intermediates. *Biochemistry*, **42**, 5713–5721.

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